Question 1: Rank and Nullity

Let

$$A = \begin{pmatrix} 2 & 1 & 4 & 3 & 0 \\ 1 & -2 & -3 & 4 & -2 \\ 0 & 1 & 2 & -1 & 1 \\ 3 & -2 & -1 & 8 & -2 \end{pmatrix} \qquad b_t = \begin{pmatrix} 2 \\ -1 \\ 1 \\ t \end{pmatrix}$$

(a) Row-echelon form and column space. Apply the indicated row operations in order:

$$\begin{aligned} A &\xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & -2 & -3 & 4 & -2 \\ 2 & 1 & 4 & 3 & 0 \\ 0 & 1 & 2 & -1 & 1 \\ 3 & -2 & -1 & 8 & -2 \end{pmatrix} \\ &\xrightarrow[R_4 \leftarrow R_4 - 3R_1]{R_2 \leftarrow R_2 - 2R_1} \begin{pmatrix} 1 & -2 & -3 & 4 & -2 \\ 0 & 5 & 10 & -5 & 4 \\ 0 & 1 & 2 & -1 & 1 \\ 0 & 4 & 8 & -4 & 4 \end{pmatrix} \\ &\xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & -2 & -3 & 4 & -2 \\ 0 & 1 & 2 & -1 & 1 \\ 0 & 5 & 10 & -5 & 4 \\ 0 & 4 & 8 & -4 & 4 \end{pmatrix} \\ &\xrightarrow[R_4 \leftarrow R_4 - 4R_2]{R_3 \leftarrow R_3 - 5R_2} \underbrace{\begin{pmatrix} 1 & -2 & -3 & 4 & -2 \\ 0 & 1 & 2 & -1 & 1 \\ 0 & 0 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}}_{\text{row-echelon form}} \end{aligned}$$

The pivot columns are columns $1, 2, 5$. Use these columns from *the original* A:

$$\mathrm{Col}(A) = \mathrm{span}\left\{ \begin{pmatrix} 2 \\ 1 \\ 0 \\ 3 \end{pmatrix} \begin{pmatrix} 1 \\ -2 \\ 1 \\ -2 \end{pmatrix} \begin{pmatrix} 0 \\ -2 \\ 1 \\ -2 \end{pmatrix} \right\}$$

These three columns form a basis because they correspond to pivots.

Question 1 (continued)

(b) Rank, nullity, and null space. First finish the row reduction:

$$R_3 \leftarrow -R_3 \quad R_1 \leftarrow R_1 + 2R_2 \quad R_2 \leftarrow R_2 - R_3$$

$$\operatorname{rref}(A) = \begin{pmatrix} 1 & 0 & 1 & 2 & 0 \\ 0 & 1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

Set $x_3 = s$ and $x_4 = q$. The equations $A\mathbf{x} = 0$ give

$$\begin{aligned} x_1 + x_3 + 2x_4 &= 0 & \qquad x_1 &= -s - 2q \\ x_2 + 2x_3 - x_4 &= 0 & \qquad x_2 &= -2s + q \\ x_5 &= 0 \end{aligned}$$

so

$$\mathbf{x} = s \begin{pmatrix} -1 \\ -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + q \begin{pmatrix} -2 \\ 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad s, q \in \mathbb{R}$$

The two vectors are independent because their third and fourth entries are $(1, 0)$ and $(0, 1)$. Hence

$$\operatorname{rank}(A) = 3 \qquad \operatorname{nullity}(A) = 2 \qquad \operatorname{rank}(A) + \operatorname{nullity}(A) = 3 + 2 = 5$$

This equals the number of columns of A, as required.

(c) Consistency and all solutions. Apply the same row operations to $[A \mid b_t]$:

$$[A \mid b_t] \longrightarrow \left(\begin{array}{ccccc|c} 1 & -2 & -3 & 4 & -2 & -1 \\ 0 & 1 & 2 & -1 & 1 & 1 \\ 0 & 0 & 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 0 & 0 & t-1 \end{array}\right)$$

The last row says $0 = t - 1$. Thus the system is consistent exactly when $t = 1$. For $t = 1$, the same remaining operations as in (b) give

$$\left(\begin{array}{ccccc|c} 1 & 0 & 1 & 2 & 0 & 1 \\ 0 & 1 & 2 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

and therefore every solution is

$$\boxed{\mathbf{x} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} + s \begin{pmatrix} -1 \\ -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + q \begin{pmatrix} -2 \\ 1 \\ 0 \\ 1 \\ 0 \end{pmatrix} \quad s, q \in \mathbb{R}}$$

Question 2: Orthogonal Projection

Let

$$\mathbf{a} = \begin{pmatrix} 1 \\ -2 \\ 2 \\ -4 \end{pmatrix} \qquad \mathbf{b} = \begin{pmatrix} 3 \\ 2 \\ 1 \\ -6 \end{pmatrix} \qquad \mathbf{y} = \begin{pmatrix} 3 \\ 2 \\ -4 \\ -1 \end{pmatrix} \qquad S = \operatorname{span}\{\mathbf{a}, \mathbf{b}\}$$

(a) Gram–Schmidt. Start with $\mathbf{a}$:

$$\mathbf{a}^\mathsf{T}\mathbf{a} = 1^2 + (-2)^2 + 2^2 + (-4)^2 = 25$$
$$\mathbf{a}^\mathsf{T}\mathbf{b} = (1)(3) + (-2)(2) + (2)(1) + (-4)(-6) = 25$$
$$\mathbf{w} = \mathbf{b} - \frac{\mathbf{a}^\mathsf{T}\mathbf{b}}{\mathbf{a}^\mathsf{T}\mathbf{a}}\mathbf{a} = \mathbf{b} - \mathbf{a} = \begin{pmatrix} 2 \\ 4 \\ -1 \\ -2 \end{pmatrix}$$
$$\mathbf{w}^\mathsf{T}\mathbf{w} = 2^2 + 4^2 + (-1)^2 + (-2)^2 = 25$$
$$\mathbf{a}^\mathsf{T}\mathbf{w} = 25 - 25 = 0$$

Both vectors have length 5, so an orthonormal basis of S is

$$\boxed{\mathbf{e}_1 = \frac{1}{5}\begin{pmatrix} 1 \\ -2 \\ 2 \\ -4 \end{pmatrix} \quad \mathbf{e}_2 = \frac{1}{5}\begin{pmatrix} 2 \\ 4 \\ -1 \\ -2 \end{pmatrix}}$$

(b) Projection and residual. Calculate the two coefficients:

$$\mathbf{y}^\mathsf{T}\mathbf{a} = (3)(1) + (2)(-2) + (-4)(2) + (-1)(-4) = -5$$
$$\mathbf{y}^\mathsf{T}\mathbf{w} = (3)(2) + (2)(4) + (-4)(-1) + (-1)(-2) = 20$$
$$\mathbf{y}^\mathsf{T}\mathbf{e}_1 = -1 \qquad \mathbf{y}^\mathsf{T}\mathbf{e}_2 = 4$$

Thus

$$\operatorname{proj}_S(\mathbf{y}) = (-1)\mathbf{e}_1 + 4\mathbf{e}_2 = -\frac{1}{5}\mathbf{a} + \frac{4}{5}\mathbf{w} = \boxed{\frac{1}{5}\begin{pmatrix} 7 \\ 18 \\ -6 \\ -4 \end{pmatrix}}$$

$$\mathbf{r} = \mathbf{y} - \operatorname{proj}_S(\mathbf{y}) = \boxed{\frac{1}{5}\begin{pmatrix} 8 \\ -8 \\ -14 \\ -1 \end{pmatrix}}$$

Check the residual against both vectors that span S:

$$\mathbf{a}^\mathsf{T}\mathbf{r} = \frac{1}{5}(8 + 16 - 28 + 4) = 0$$
$$\mathbf{b}^\mathsf{T}\mathbf{r} = \frac{1}{5}(24 - 16 - 14 + 6) = 0$$

Therefore $\mathbf{r}$ is orthogonal to all of S.

(c) Distance. The distance is the length of the residual:

$$\operatorname{dist}(\mathbf{y}, S) = \|\mathbf{r}\| = \sqrt{\frac{8^2 + (-8)^2 + (-14)^2 + (-1)^2}{5^2}} = \sqrt{\frac{325}{25}} = \boxed{\sqrt{13}}$$

Question 3: Recommending Content on an Online Platform

Write $\mathbf{u}_i \in \mathbb{R}^{d_u}$ for user i and $\mathbf{v}_j \in \mathbb{R}^{d_v}$ for video j, where $d_u \neq d_v$. Let n_u, n_v be the numbers of users and videos.

(a) Clustering. Cluster the video vectors into K groups. Let $\boldsymbol{\mu}_k \in \mathbb{R}^{d_v}$ be the centre of group k. In preprocessing, keep a small bank C_k of at most B representative videos from each group. A user is first matched to a few promising groups, and only videos in their banks are scored. This avoids comparing that user with all n_v videos. Different groups also provide different kinds of videos.

(b) Matrix score. For nonzero vectors, define

$$\widehat{\mathbf{u}}_i = \frac{\mathbf{u}_i}{\|\mathbf{u}_i\|} \in \mathbb{R}^{d_u} \qquad \widehat{\mathbf{v}}_j = \frac{\mathbf{v}_j}{\|\mathbf{v}_j\|} \in \mathbb{R}^{d_v}$$

$$W \in \mathbb{R}^{d_u \times d_v} \qquad s(i,j) = \widehat{\mathbf{u}}_i^{\mathsf{T}} W \widehat{\mathbf{v}}_j \in \mathbb{R}$$

The matrix W connects coordinates with different meanings and dimensions. A direct dot product $\mathbf{u}_i^{\mathsf{T}}\mathbf{v}_j$ is not defined when $d_u \neq d_v$. Normalising the vectors removes differences in their lengths, so the score focuses on the learned match. Rank videos by larger $s(i,j)$.

(c) Supervised learning and fast recommendations. Use one training example for each *viewed* user–video pair (i,j). Its features are $(\widehat{\mathbf{u}}_i, \widehat{\mathbf{v}}_j)$, and its label is one of L (like), D (dislike), or N (viewed with no response). An unviewed pair is not labelled N.

Learn three matrices and three scalar biases:

$$M_L, M_D, M_N \in \mathbb{R}^{d_u \times d_v} \qquad b_L, b_D, b_N \in \mathbb{R}$$

$$z_c(i,j) = \widehat{\mathbf{u}}_i^{\mathsf{T}} M_c \widehat{\mathbf{v}}_j + b_c \qquad c \in \{L, D, N\}$$

$$p_c(i,j) = \frac{e^{z_c(i,j)}}{e^{z_L(i,j)} + e^{z_D(i,j)} + e^{z_N(i,j)}}$$

For the set T of viewed pairs, with observed label y_{ij}, minimise

$$\mathcal{L}(M_L, M_D, M_N, b_L, b_D, b_N) = -\frac{1}{|T|} \sum_{(i,j)\in T} \log p_{y_{ij}}(i,j) + \lambda \sum_{c\in\{L,D,N\}} \|M_c\|_F^2 \quad \lambda \geq 0$$

Here $\|M_c\|_F^2$ is the sum of the squares of all entries of M_c. After training, set $W = M_L - M_D$ in part (b). Then

$$\begin{aligned} \log \frac{p_L(i,j)}{p_D(i,j)} &= z_L(i,j) - z_D(i,j) \\ &= \widehat{\mathbf{u}}_i^{\mathsf{T}} W \widehat{\mathbf{v}}_j + (b_L - b_D) \\ &= s(i,j) + (b_L - b_D) \end{aligned}$$

The bias difference is the same for every video of one user. Thus ranking by $s(i,j)$ is exactly ranking by the model's like-to-dislike odds.

Question 3 (continued)

For each centre, normalise $\boldsymbol{\mu}_k$ and precompute $W\widehat{\boldsymbol{\mu}}_k$. Also precompute $W\widehat{\mathbf{v}}_j$ for every video in a bank. For a user, rank the K centres by

$$s(i,k) = \widehat{\mathbf{u}}_i^{\mathsf{T}} W \widehat{\boldsymbol{\mu}}_k$$

Choose the best h groups, score at most hB bank videos with $s(i,j)$, and return the best ℓ videos, where $\ell \le hB$. The banks may be chosen by popularity and diversity within each cluster. This is approximate because some videos are absent from the banks.

Assume K, B, h, ℓ are fixed as n_v grows, $h \le K$, and the embeddings, centres, banks, and products with W are prepared in advance. With fixed embedding dimensions, the time per user is

$$O(d_u K + K \log h + d_u hB + hB \log \ell)$$

which has no n_v term. If all user and video embeddings and each video's group number are kept, memory is

$$O(n_u d_u + n_v d_v + n_v + K d_v + KB(d_v + d_u) + K d_u + 3 d_u d_v)$$

Here $KB(d_v + d_u)$ stores the bank videos and their precomputed products. If the $|T|$ training records are also kept, add $O(|T|)$ memory. Small extra arrays for biases and the output list are included in the bound.

(d) Drawback. Only videos in the selected banks can be recommended. A new or less popular video may be missing even when it would suit a user, so the system can miss good recommendations.

Question 4: Matrix Products and Rank

Let $A \in \mathbb{R}^{m \times n}$.

(a) Show $\operatorname{rank}(A^\mathsf{T}A) = \operatorname{rank}(A)$**.** First, if $A\mathbf{x} = 0$, then

$$A^\mathsf{T}A\mathbf{x} = A^\mathsf{T}0 = 0$$

Conversely, suppose $A^\mathsf{T}A\mathbf{x} = 0$. Multiply on the left by $\mathbf{x}^\mathsf{T}$:

$$\begin{aligned} 0 = \mathbf{x}^\mathsf{T}A^\mathsf{T}A\mathbf{x} = (A\mathbf{x})^\mathsf{T}(A\mathbf{x}) = \|A\mathbf{x}\|^2 \\ = \sum_{r=1}^{m} (A\mathbf{x})_r^2 \end{aligned}$$

Every term is nonnegative, so every term is zero. Hence $A\mathbf{x} = 0$. We have shown

$$\operatorname{Nul}(A^\mathsf{T}A) = \operatorname{Nul}(A)$$

Both matrices have n columns. Apply rank–nullity to each:

$$\begin{aligned} \operatorname{rank}(A^\mathsf{T}A) &= n - \operatorname{nullity}(A^\mathsf{T}A) \\ &= n - \operatorname{nullity}(A) \\ &= \boxed{\operatorname{rank}(A)} \end{aligned}$$

(b) Invertibility and independent columns. The matrix $A^\mathsf{T}A$ is $n \times n$. Thus

$$\begin{aligned} A^\mathsf{T}A \text{ is invertible} &\iff \operatorname{rank}(A^\mathsf{T}A) = n \\ &\iff \operatorname{rank}(A) = n \\ &\iff \text{the } n \text{ columns of } A \text{ are linearly independent} \end{aligned}$$